Analysis of Data Research in Smart GIS Central Database using Map Reduce Pattern and Bayesian Probability

Moubaric KABORE[1], Abdoulaye SERE[2], Vini Yves Bernadin LOYARA[3]

[1,2]Equipe de Recherche Signal, Image et Communications (ER-SIC),
Laboratoire d'Algèbre de Mathématiques Discrètes et Informatique (LAMDI),
Université Nazi BONI, Burkina Faso
[1,2]Réseau des Enseignants Chercheurs et Chercheurs en Informatique du Faso (RECIF), https://recifaso.org
[1]kmoubaric24@gmail.com,
[2]abdoulaye.sere@recifaso.org,
loyarayves@yahoo.fr,

Abstract. This paper deals with Bayesian approach in Data Research in GIS database through artificial Intelligence (AI) modules, reading the best bayesian probability before returning the data requested, denoted AI4DB. The proposed method combines meshing techniques and the map-reduce algorithm with Bayesian approach to obtain an intelligent GIS database to reduce the execution time of all the entiere system. According to the values of the Bayesian probability, the nearest sites of any position resulting of the user requests, are extracted speedily from the database using the map reduce framework. The proposed method is applied to vehicle research in sellers in the cities of Ouagadougou and Bobo Dioulasso. the execution time is less than the time for the case of the classical method, based only on a parallelism research in a fixed virtual grid without probability. Only a map function with the best bayesian probability for the data in entry, executes entirely its instruction.

Keywords: Machine Learning · MapReduce · GIS Database · Bayesian Probability · NOSQL database

1 Introduction

Today, technologies are evolving speedily with innovations introduced by the Internet Of Things (IOT), Big Data, and Artificial Intelligence. According to International Telecommunication Union (ITU), the Internet Of Things (IOT) is the global infrastructure for the information society, enabling advanced services by interconnecting (physical and virtual) things based on existing and evolving interoperable information and communication technologies.

The number of things or object connected in the world is inscreasing each year, according to International Telecommunication Union (ITU) statistics : the

Internet of Things engages the eve of a new revolution where the physical object or the virtual object are connected and can discuss between each other.

The Internet Of Things (IOT) is essential in Big data collection, because data collection is facilitated by connected objects : 4G and 5G also contribute to data collection. A large amounts of data is necessary to achieve best decision making for industries and commercial enterprises. Multiple sensors, social networks and applications on mobile phones have contributed strongly to data collection, even in streamming mode. For instance, Sere and others in [12] have proposed classification of posts from social networks to collect, to analyze speedily Big data in streaming mode with spark tools.

Big data technologies proposes multiple tools to capture, to store, to analyze and to visualize data. Nosql, newsql databases offer capacities to store big data and accept connection from data processing tools such as Hadoop, spark that bring out functionalities through available API related to Machine Learning and map-reduce framework. For instance performance benchmarking of nosql database has been studied by Irina Astrova and others in [14], to present a view on the best solutions between nosql databases and newsql database for data storage and performance.

Many tools related to the big data processing, connected on nosql database, propose librairies for Artificial intelligen ce (AI). If the layer i does not take Artificial intelligence into account, the layer i+1 will supply the problem by introduction inside Artificial intelligence techniques. For example, spark proposes AI libraries that could be used to make intelligent nosql databases. Cloud computing also offers opportunities of services and infrastructures that strenghten and facilitate administration digitalization, giving possibilities to use Big data technologies in the cloud, to reduce financial resources used to deploy and to maintain infrastructure, to support human resource training for innovations and to compensate for the lack of competence in the case of local service. In this way, Didier and others in [17] have studied the challenges and the opportunities of Gcloud engaged by the governement of Burkina Faso to enhance ICT economy.

Scientists have worked to develop algorithms to analyze Big Data. For instance, the map-reduce algorithm, initially introduced by Dean and others in [7] to count the words, has been extended to recognize features in an image in [2], to extract posts in social networks in [12] and to research GIS Data in [16] by SERE and others.

Data research in big data to get speed responses is a challenge that face governements to enhance administration services. Optimization of each thread instruction reduce its execution time in the context of application Scability.

All the fields of computer science are concerned with artificial intelligence (AI) as Database, image processing, Interface Human and Machine (IHM), network and software security. Scientists have tried to apply AI to classical problems in any field of computer science. The question is how to use artificial intelligence techniques in these fields to improve classical methods about efficacy in taking decision. Database and artificial intelligence (AI) can help each other accordingly to Zhou and others in [19]. AI can transform database to be more intel-

ligent(AI4DB). That means, an intelligent database is a database which uses the techniques of artificial intelligence to response to unpredictable events to the request of users, to return relevant data in a short time instead of using mechanicaly manipulation languages. Face to scability, an intelligent database could propose alternatives to reduce its impacts to predict applications behaviour. Yue Dong and others in [18] have explained the importance of intelligent database in enginering application. On the other hand, data from database could be used to optimize AI algorithm, denoted (DB4AI).

Several works have been proposed by scientists to analyze the application of bayesian probability to the GIS data, as presented by Ames and others in [23] to explain the cases of Bayesian probablity integration in GIS. Stassopoulou and others in [22] have demonstrated their approach of bayesian networks within the framework of the problem of assessing the risk of desertification of some burned forests in the Mediterranean region. Junpeng Huang and others in [20] have used bayesian networks to study landslides that frequently occur ccur along the eastern margin of the Tibetan Plateau, which poses a risk to the construction, maintenance, and transportation of the proposed Dujiangyan city to Siguniang Mountain (DS) railway, China. Galan and others in [21] have studied an application of bayesian networks in reforestation planning : they have constructed a reforestation model for woodland located in the basin of the river Liébana (NW Spain).

How to reduce the time of data research in GIS database to improve the time necessary to response to the request services from customers, in looking for data in spatial data? In this sense, in more datails, how to combine parallelism through threads, and prediction techniques to the reduce the time attributed to data research in GIS database?

Our motivation is focusing on improvements of data research in GIS database, through the initial works based on meshing techniques and the map-reduce algorithm, introduced by sere and others in [16], to explore Machine Learning Technique with application of probability technique, as Bayesian approach.

Therefore, this paper is organized as follows : section 2 recalls problem statement, bayesian probability definition and ends with the map-reduce algorithm. The introduction of bayesian approach is also discussed in application in our method in section 3. Experimental analysis is illustrated in section 4 and explains comparisons between the results of the proposed method and the framework for data research in GIS database, without using bayesian approach, introduced by sere and others in [16], to highlight different improvements in each case.

2 Preliminaries

This section concerns with the problem statement, described in section 2.1. It also recalls the map-reduce algorithm and the classical bayesian approach in Machine Learning (ML) that will contribute to perform data research in order to achieve a smart GIS database.

2.1 Problem statement

The problem statement is an extension and a continuation of the problem statement introduced by Sere and others in [16] about data research in GIS Database, but slightly modified. Let W be the universal set of database sites distributed in a grid, named G superimposed on the image space. Let S be a subset of database sites such as $S \subset W$. Suppose that $S = \{S_1, S_2, \ldots, S_{n-1}, S_n\}$ where S_i is a database site. All the data in the database sites $S_i \in$ S together leads to the characteristics of Big Data, especially through data volumetric.

S contains a large amounts of data such as Data research in S needs implementation of technologies for big data. Using threads is a well known alternative to process speedily and simultaneously each subset of big data.

Data research in S is based on a set of successives conditions or criteria. Suppose that $C = \{c_1, c_2, \ldots, c_{k-1}, c_k\}$, this set of criteria. C describes the constraints that the extracted data from the set S by users requests must respect.

Consider d as a specific data such as $d \in u$ and $u \in W$. u could be a member of S or not. Let M be a point. The problem is to find out the database sites $S_i \in S$ where the database sites S_i must be the neighbors of the point M and effectively with $d \in S_i$.

For example, the nearest neighbors of the point M is the criteria, represented by $c_v \in C$.

For c_v, there are two manners to verify if the data d is in the database S_i. That means to response to the question if $d \in S_i$? :

- the first one is to search data sequencialy in each $S_i \in S$ in going from 1 to n.
- the second one is to proceed by a parallel (or to mix sequential and parallel) verification with research in each group of S_i, using for instance probability approach as bayesian probability.

Our hypothese is that the parallel verification is faster than the first one, even faster with introduction of bayesian probability criteria for data research.

Suppose that α is the time taken to perform S_i.

In the first case, the effective execution time will be $n\alpha$ in the worst case. The worst case corresponds to the conditions (if $d \in S_n$ or if d is not a member of any S_i).

In the second case, let β be the number of group of S_i. That means clearly $\beta \leq n$. θ will be also considered as the execution time to perform each group. The global execution time will be $\beta\theta$. We obtain $\beta\theta \leq n\theta \leq n\alpha$ because $\beta \leq n$.

Bayesian approach introduces the probability to verify if $d \in S_i$. It allows to begin data research with clusters or database sites, having best Bayesian probability, as a priority criteria to classify clusters or database sites.

Suppose that $c_b \in C$, the bayesian criteria. Our hypothese is that c_b will contribute to have the relation $\gamma \leq \beta$ and then $\gamma\theta \leq \beta\theta$, to enhance the parallel verification time.

2.2 Mesh Generation

Mesh meets applications in image processing to represent pixels or others geometric shapes in a grid to allows algorithm implementation, optimization of site localisation in a grid and Data reseach in GIS database, as presented by sere and others in [16].

Here, a virtual grid is introduced on the image to delimite clusters that contains database sites, as discussed in [16] to define clusters.

Several techniques of creating virtual grids exist. For instance, quasi affine transformation definition is presented by Marie-Andree and others in [9] and allows to obtain a regular grid. Voronoi Diagram and Delaunay Triangulation give anothers methods to create virtual grids.

The technique of deploying a grid depends on the distribution of database sites. Antoine Vacavant'thesis in [24] exposes a variant of regular grids and irregular grids that could be studied to create clusters, in more details in the future works. Probability of different clusters will be evaluated by threads in background process in the DBMS. For instance, bayesian approach, stochastic process and geostatistics can be explored in application to the clusters or the database sites in the clusters.

Here, neighborhood is based on 8-neighborhood, as described in [1] and in [8]. 8-neighborhood uses the notion of distance and determines the nearest sites.

2.3 Map-reduce analysis

The Map-Reduce algorithm is a technique based on the parallelism execution of data, creating different threads to process speedily splitted big data, as explained initialy by Dean and others in [7] and applied by SERE and others in[12] to extract posts in the social networks. The map reduce algorithm consists of three phases as the map function, The shuffle phase followed by the reduce phase.

The map reduce model is a conbinaison of successive famillies of three functions formally noticed by Reduce(Shuffe(Map (data))) or Reduce o shuffle o map(data) applied to big data processing. The map function application considers that big data processing needs firstly speed transformations to collect, to filter, to select relevant data. Secondly, before storing data on database, data simplification and data classification through the shuffle phase facilitate forward next operations. Thirdly the reduction phase is then applied to each class to transforms its value.

The map-reduce framework has been used to resolve several problems such as counting the number of word in a document by Dean and others in [7], computing the average of numbers, making data selection and sorting a dataset.

In more details, the map function transforms as inputs a single key and a value, noticed (k, v) pair and produces as outputs a set of intermediate (key, value) pairs (k_i, v_i). All the map functions are executed simultanouesly. Finally, the map function is just a transformation and could contain any instruction. For example, the map function takes the pair (k, v) as an input and produces the set of pairs (k_1, v_1), (k_2, v_2). Thus, several pairs (k_i, v_i) are produced by all the

map functions together. In the next sections, the idea is to inject instructions to deal with Machine Learning Techniques, into the map function to improve data research.

The shuffle phase is an intermediate phase between the map phase and the reduce phase. Therefore, it starts after all the map functions ended and before starting the reduce functions. The shuffle phase plays the role of data classification : it puts together the value of pairs having the same key. For example, for the following pairs $(k_1, v_1), (k_1, v_2), (k_2, v_2), (k_2, v_3)$ generated by all the map functions, the shuffle phase returns the pairs $(k_1, < v_1, v_2 >), (k_2, < v_2, v_3 >)$. Thus, the shuffle phase carry out data classification where each class is referenced by its key.

The reduce function takes as an input a (key, list of values) pair that contains a intermediate key and a set of values for that key. The reduce function produces a pair (key, result of a list of values). The key in the input is the same in the ouput. For instance, the pairs $(k_1, < v_1, v_2 >)$ and $(k_2, < v_2, v_3 >)$ become $(k_1, < v_1 + v_2 >), (k_2, , < v_2 + v_3 >)$.

Reduce functions could start before the end of all the map functions. Mixing map and reduce should reach probably an improvement of processing time : That will be studied in more details in perspectives, in considering sychronization control between map, reduce and shuffle.

The map-reduce model consists of the parallelism of map function followed respectively by the shuffle phase and the reduce functions : there is a master node that controls all the processus tasks, distributed on secondary nodes with distributed memories.

Thus, the map function, the shuffle phase and the reduce function have summarized successively in 1, 2 3, 4, 5, 6 and 7 as follows :

$$Map(k, v) \longrightarrow \{(k_1, v_1), (k_1, v_2)\} \tag{1}$$

$$Map(k', v') \longrightarrow \{(k_2, v_2), (k_2, v_3)\} \tag{2}$$

$$\{(k_1, v_1), (k_1, v_2)\} \longrightarrow_{shuffle} \longrightarrow (k_1, < v_1, v_2 >) \tag{3}$$

$$\{(k_2, v_2), (k_2, v_3)\} \longrightarrow_{shuffle} \longrightarrow (k_2, < v_2, v_3 >) \tag{4}$$

$$Reduce(k_1, < v_1, v_2 >) \longrightarrow (k_1, < v_1 + v_2 >) \tag{5}$$

$$Reduce(k_2, < v_2, v_3 >) \longrightarrow (k_2, < v_2 + v_3 >) \tag{6}$$

Generally the reduce function is defined by :

$$Reduce(k_i, < v_1, ..., v_j >) \longrightarrow (k_i, v_k) \tag{7}$$

Where $v_k = v_1 + ... + v_j$, being the result of the operator + applied to the members of the list $< v_1, ..., v_j >$.

Forward, the symbols M, S, R shall correspond respectively to the functions Map (M), Shuffle (S) and Reduce (R).

Prediction is introduced in data research throught the presence of a specific vehicle at a seller, using bayesian probability.

2.4 Bayesian Probability Description

In probability theory, Bayes 's theorem is based on the conditionnal probability. Bayes's probability is defined by :

Definition 1. *(BAYES formula) For any pair of events A, B, the conditional probability formula becomes the following formula called Bayes's formula :*

$$\mathbb{P}\left(A/B\right) = \frac{\mathbb{P}\left(A\right) \times \mathbb{P}\left(B/A\right)}{\mathbb{P}\left(A\right) \times \mathbb{P}\left(B/A\right) + \mathbb{P}\left(\bar{A}\right) \times \mathbb{P}\left(B/\bar{A}\right)} \tag{8}$$

The term $\mathbb{P}\left(B\right) \times \mathbb{P}\left(B/A\right)$ is called the likelihood function of B. $\bar{A}$ is the complement of A.

More generally, if A_i is a quasi-complete system of finite or countable events,

$$\mathbb{P}\left(A_i/B\right) = \frac{\mathbb{P}\left(B/A_i\right) \times \mathbb{P}\left(A_i\right)}{\sum\limits_j \left[\mathbb{P}\left(B/A_j\right) \times \mathbb{P}\left(A_j\right)\right]} \tag{9}$$

for any A_i event of the system.

Let A and B be two events. Bayes's theorem allows to determine the probability of A knowing B, denoted $\mathbb{P}\left(A/B\right)$, as follows :

- let $\mathbb{P}\left(A\right)$, $\mathbb{P}\left(B\right)$, $\mathbb{P}\left(A/B\right)$, $\mathbb{P}\left(A \cap B\right)$ be respectively the probability of A, the probability of B, the probability of A knowing B, the probability of A and B both occur. The conditional probability is then defined by :

$$\mathbb{P}_B\left(A\right) = \mathbb{P}\left(A/B\right) = \frac{\mathbb{P}\left(A \cap B\right)}{\mathbb{P}\left(B\right)} \tag{10}$$

The following formulas follow immediately the conditional probability.

- If $A, B \in \mathcal{A}$;

$$\mathbb{P}(A \bigcap B) = \mathbb{P}(B) \times \mathbb{P}(A/B) \tag{11}$$

- If $A, B, C \in \mathcal{A}$;

$$\mathbb{P}(A \bigcap B \bigcap C) = \mathbb{P}(B) \times \mathbb{P}(A/B) \times \mathbb{P}(C/(A \bigcap B)) \tag{12}$$

More generally if $A_1, A_2, \cdots, A_n \in \mathcal{A}$, then

$$\mathbb{P}\left(A_1 \bigcap A_2 \bigcap \cdots \bigcap A_n\right) = \mathbb{P}\left(A_2\right) \times P\left(A_1/A_2\right) \times \mathbb{P}\left(A_3/(A_1 \bigcap A_2)\right) \times \cdots \times \mathbb{P}\left(A_n/(\bigcap_{k=0}^{n-1} A_k)\right) \tag{13}$$

More details on Bayesian probability are presented in the next sections. Bayesian probability is integrated into the tables of database , respectively with its corresponding value for a cluster or for a site localized in a cluster, in order to improve the execution time of data research to response to the requests.

3 Method description

In software engineering, the Model-View-Controller (denoted MVC) is very used in component structuration, rendering easy code maintenance. As illustrated in figure 1, MVC presents an updating data process and a reading data process that concerns our study. Udapting data process consists of threads in background, to synchronize data from database sites.

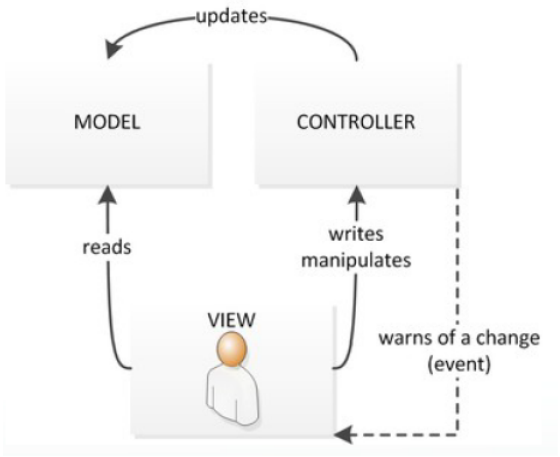

Fig. 1. Model-View-Controller (denoted MVC)

This section describes the model component, through a global architecture presented in figure 2 in section 3.1 and the content of the map-reduce algorithm (for instance localized inside or outside of the database with an implementation in spark), reading data in taking into account the bayesian probability of data. It also explains how to introduce bayesian approach application in data research in GIS database. The database saves regularly both essential data and its corresponding bayesian probability in others tables.

Here, the content of the database and its internal associated algorithm performs the data as for the description of an artificial assistant in the real case to determine the best response. Bayesian probability plays the rule of this artificial assistant.

3.1 Global architecture

The idea is to introduce an assistant in the database to give orientations, advices or indications on how to begin data research in specified clusters. In this sense, there are three levels of analysis :

For data research, a global architecture using a deployment diagram defined in Unified Modelling language (UML), is presented in figure 2 :

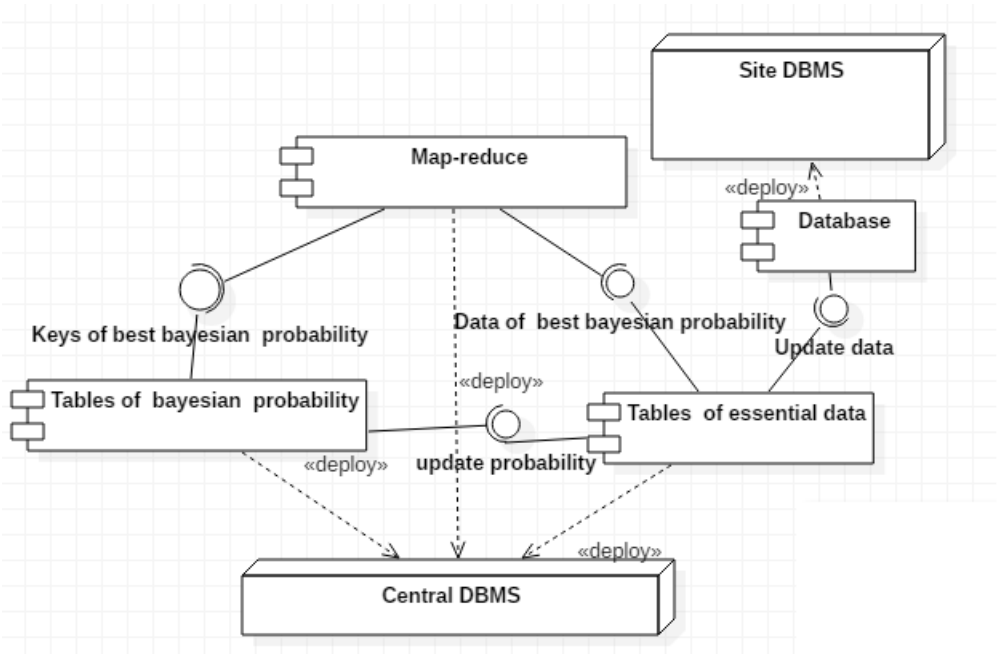

Fig. 2. the global architecture using Unified Modelling language (UML)

This architecture shows several entities :

- the map-reduce entities in spark : this entity analyzes the contain of each tuple to read probability value before emitting the final tuple.
- the tables of essential data in DBMS : the database represents different table with essential data for business intelligence. Synchronisation of these data is realized regularly with the data from different database sites in a background process.
- the tables of data with bayesian probability in DBMS related to tables for bayesian probability managment by threads in background process that makes smart the database : it facilitates and prepares data research with udapting probability in order to improve and to shorten the execution time to search a specific item in the database.

3.2 Application of bayesian approach

Each database site is associated to a probability, defined by bayesian approach to find inside an item at a given moment t_i. Bayesian Probability is updated regularly by a process running in background at each interval δt : that means at each t_i+δt, bayesian probability of data is computed and stored to follow and to know probability tendency.The value of δt must be a parameter fot the system. Each localized database for each seller is synchronized with the unified database of the system to udapte regularly different transactions related to retry vehicles or to add new vehicles. A process running in background as threads updates bayesian probability of any data, corresponding to vehicle descriptions for each cluster and for each site (in a cluster). Optimization of each thread can influe globaly on scability. One of the question is the order of each condition application to the data and how to manage easily scability.

Let A_i and B be respectively a seller and a vehicle to research in the database. The set of A_i constitutes an almost complete system of events. The probability of finding a vehicle B knowing that it comes from the seller A_i was defined by the relation 9 and will lead to the following formula :

$$\mathbb{P}\left(B/A_i\right) = \frac{\mathbb{P}\left(A_i/B\right) \times \sum_j \left[\mathbb{P}\left(A_j/B\right) \times \mathbb{P}\left(A_j\right)\right]}{\mathbb{P}\left(A_i\right)} \tag{14}$$

A vehicle B is in three sellers localised in the same neighborhood, A_1, A_2 and A_3, respectively the seller A_1 has an average of 0.3% of B ; A_2 has 0.8% and A_3 has 1% . The three sellers have together one thousand (1000) vehicles of B in stock, distributed as follows : five hundred (500) in the stock of the seller A_1, three hundred fifty (350) in A_2 and one hundred fifty (150) in A_3.

Consider, someone wants to buy the vehicle B. What is the probability that the vehicle B comes from the seller A_1 (or A_2 or A_3)?

We have :

$$P\left(A_1\right) = 0,5,\ P\left(A_2\right) = 0,35,\ P\left(A_3\right) = 0,15 \tag{15}$$

The problem is to determine the conditional probabilities :

$$P\left(A_1/B\right),\ P\left(A_2/B\right),\ P\left(A_3/B\right)$$

We know that $P\left(B/A_1\right) = 0,003$, $P\left(B/A_2\right) = 0,008$ and $P\left(B/A_3\right) = 0,01$. Then

$$P\left(A_1/B\right) = \frac{P\left(B/A_1\right) \times P\left(A_1\right)}{P\left(B/A_1\right) \times P\left(A_1\right) + P\left(B/A_2\right) \times P\left(A_2\right) + P\left(B/A_3\right) \times P\left(A_3\right)} = 0,26 \tag{16}$$

In the same way, we get

$$P\left(A_2/B\right) = 0,48,\ P\left(A_3/B\right) = 0,26 \tag{17}$$

Probability brings out another condition in data processing in additional to the nearest sites, defined by 8-neighborhood due to mesh generation, explained previously by SERE and others in [16].

As presented, in problem statement, the list of criteria, as constraints is represented by $C = \{c_1, c_2, \ldots, c_{k-1}, c_k\}$. A memory table T_i is created and associated to each constraint c_i : the set C leads to k memory tables.

Finally, the GIS database will contain three entities :

- the GIS table contains the data associated to the GIS, represented by the table 1.
- the region table shows only the region name and its image, represented by the table 2.
- k intermediate memory tables are introduced in our analysis, corresponding to k constraints defined by the set $C = \{c_1, c_2, \ldots, c_{k-1}, c_k\}$. Each memory table role is to facilitate changing probalility and to shorten the execution time of data research in the GIS Database. Memory tables make GIS database more intelligent. An example of memory tables using bayesian probability on data is presented in different tables 3 and 4 . The data in these tables is sorted according to the biggest bayesian probability of the data name on the cluster and then the biggest bayesian probability of the data name on the site in the same cluster.

Table 1. Table of main data (GIS table)

Region name	Cluster Ref	Site Ref	Data name
secteur 1	(0, 0)	(0,1)	Data on the site
secteur 1	(0,0)	(0,1)	Data on the site
secteur 2	(2,2)	(0,4)	Data on the site
secteur 2	(2,2)	(0,4)	Data on the site
secteur 2	(2,2)	(0,4)	Data on the site

Table 2. Table of regions

Region name	images
secteur 1	image blob

Table 3. A memory table for any data, where BP means Bayesian Probability

Cluster reference	BP on the cluster	Data name
(0,0)	0.5	Data
(0,0)	0.5	Data

Table 4. A memory table for any data, where BP means Bayesian Probability

Site Ref	Data name	BP on the site
(0,1)	Data on the site	0.6
(0,1)	Data on the site	0.6

The application of a constraint to the data in the GIS database is a projection on the data. Before going to look for data directely in the GIS table, the table T_i contributes to reduce the execution time for data research operation in the GIS table : Verification in the tables T_i return the list of items to research in the GIS table.

All the tables T_i are updated automatically, in real time, with the adequate bayesian probability only by the Database Management System (DBMS).

But, through application services, the users could create the table with its attributes and choice probability available by the library of DBMS.

Moreover, most of the Database Management System (DBMS) propose a language, to write scripts to create tables, procedures or functions and to compute baeysian probability in a process working in background, for prediction based on data in the tables.

If the number of memory tables denoted card(C) is increasing, the time taken to manage these tables will take more time. Therefore, the users will receive tardily the responses related to data research.

Sere and others in [16] have proposed a solution to read successively data in the order in the neighbor clusters, following 8-neighborhood. When reading a cluster, its database sites are read sequentially.

But, here each cluster is associated to its bayesian probability : reading data will begin with the cluster that has the biggest bayesian probability. In the same cluster, the database sites have read in following the order of the biggest bayesian probability associated to the sites.

Algorithm 1 presents the function UpdateTable that defines the procedure to update regularly bayesian probability in the tables T_i. The instructions of Algorithm 1 will be executed regularly as a thread, implemented as a trigger in a background process by the DBMS itself : it consist of verifying new records in the tables and to compute bayesian probability. It could be in parallelism with all the threads, responsible of the map-reduce algorithm, called by the users.

Algorithme 1 : function UpdateTable(Listofclusters liste)

```
Result : the list of clusters and its internal sites are updated
c : Cluster ;
for each cluster c in liste do
  | updateclusterandsites(c);
end
```

The proposed method also gives possibilities to use together different probability on the tables T_i or doing comparisons on the execution time separately with each probability. These cases of comparisons could be studied in more details in the future works.

3.3 Application of map-reduce pattern

The initial structure of Reduce(Shulle(Map)) named R o S o M with three layers is adapted to the wordCount technique proposed by Dean and others in [7].

In this way, suppose that :

$$P = \{SoRoM, SoMoR, RoMoS, RoSoM, MoRoS, MoSoR\} \tag{18}$$

P represents the set of compositions generated by the layers R, M, S together and contains entirely six members.

The question is why not used the members of P, accordinly to the data in the entries of the database and to force only focusing on the classical model R o S o M, adapted to the wordcount from Dean and others in [7] ?

Moreover, the general formula to process Big data could be defined by the combinaison of elements in the relation R=$\{R_1, R_2, ..., R_{(i-1)}, R_i\}$. That means Each R_i consists of the parallel execution of data or the parallel execution of instructions. These R_i could be considered as tasks, synhronized or not, represented by a graph that leads to others alternatives, to be studied in more details in the future works.

But, here, the content of the map-reduce algorithm proposed in [16], has been updated to take bayesian approach into account as follows :

- Algorithm 2 presents the map function. It uses the function verification to read the tables T_i of baysesian probability, before reading the GIS table, to return right informations. The data in the table is classified in ascendant order to facilitate data research, accordingly to the big probability of each cluster. The content of the function verification is illustrated in algorithm 3 : it consists of looking for the probability of (idcluster, value) among the best probability ;
- algorithm 4 describes the content of the reduce function.

Algorithme 2 : function map(Doc idcluster, Doc value)

```
Result : the pairs in the form (k, v)
d : Data ;
information : String ;
if Verification(idcluster, value ) then
    information←getInformationSite(value) ;
    emit(idcluster, information);
end
```

The function verification is defined in algorithm 3.

Algorithm 3 describes the content of the function verification to determine the correspondance of the best probability to the selected cluster.

The next section will give details on implementations of these algorithms.

Algorithme 3 : function verification(Data d, Doc value)

```
Result : the pairs in the form (k, v)
d : Data ;
information : String ;
if bestProbability(idcluster, value ) then
    return true;
    else
    return false;
end
```

Algorithme 4 : function reduce(Docid idCluster, Iterator value)

```
Result : the pairs in the form (k, < v_1, v_2, ..., v_{n-1}, v_n >)
result : String ;
result ← "";
for each v_i in value do
    result= result+" "+v ;
end
emit(result);
```

4 Illustrations and Discussions : the case of vehicle sellers in Burkina Faso

Simulation illustrates the description of the dataset and the application of the proposed method to GIS data, with the content of map-reduce initially defined.

4.1 Datasets

Since march 03, 2022 to march 08, 2022, we collected the data with KoBotoolbox at the sellers of cars, vans and trucks in Ouagadougou and in Bobo-Dioulasso , the two largest cities in Burkina Faso. For instance, Figure 3 and 4 show the regions where the data have been collected, respectively the data at the sellers in Ouagadougou and in Bobo Dioulasso (the study area) : The corresponding points of sellers are repeated on the same points, according to the scale taken and applied to the image.

The data describe the characteritics of cars as the name, the types of vehicles, the GPS of the seller, the numbers of selled vehicles. Table xxx presents a sample of these collected data. The data have been adaptated and loaded in the mongodb database, as a set of documents in a collection, presented in the next section 4.2.

The files KML of the previous two maps in figures 3 and 4, have been exported from KoBotoolbox to use in a program in python for easy manipulation distance and meshing.

Moreover, the global (or local) convex hull of sites (points) is respectively the region where any customer could meet easily all the sellers (or the sellers in a region) of cars, vans and trucks. Most of the sellers are aligned on the main

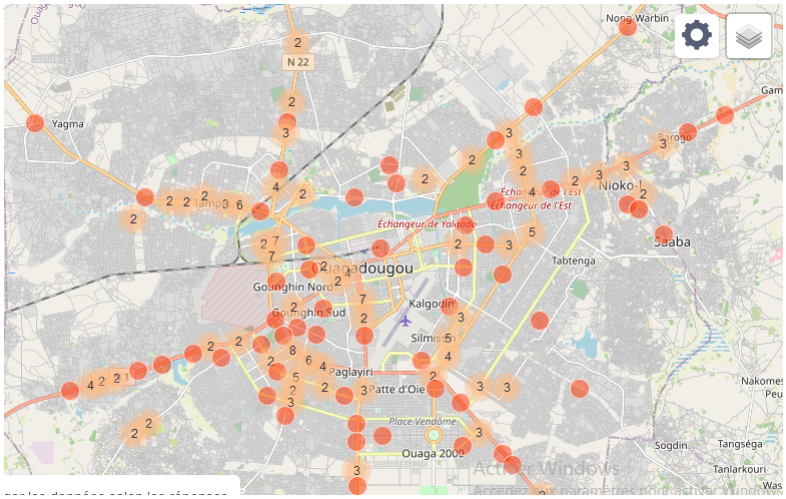

Fig. 3. one hundred three (103) repeated sellers on the same points of cars, vans and trucks in Ouagadougou, Burkina Faso

Fig. 4. sixty four (64) repeated sellers on the same points of cars, vans and trucks in Bobo Dioulasso, Burkina Faso

roads in the Figures (3 and 4) : the modelization of roads by a graph or a tree with the data in the nodes. In this case, data research consists of searching data in a graph. The future works will focus on research in a graph, implemented in a GIS database, precisely in a graph NOSQL database as an alternative.

4.2 Discussions

Consider $S = \{S_1, S_2, \ldots, S_{n-1}, S_n\}$ a set of sellers, corresponding to the database sites distributed in different clusters in a region. Suppose that all the data size corresponds to volumetric, as one of the characteristics of Big Data. Clustering allows to build separated groups of data in order to apply the map function to each group for data analysis.the The map reduce framework also facilitates data research with introduction of the parallel execution in the entry data, extracted from the clusters.

If k=2, the previous set of constraints $C = \{c_1, c_2, \ldots, c_{k-1}, c_k\}$ will become $C = \{c_1, c_2\}$. For instance, c_1 corresponds to the constraint related to the nearest database sites defined by 8-neighbourhood, the algorithm defined by sere and others in [16]. In the sense of the contraint c_1, suppose S_1, S_2 and S_3 are only the database sites, localized in three clusters in 8-neighbourhood : S_1, S_2 and S_3 are respectively in the clusters abstracted and referenced by (0,0), (0, 1), (1, 0). While c_2 indicates the bayesian probability of the vehicle "toyota". c_2 leads to two cases of bayesian probability, named $c_{2,1}$ and $c_{2,2}$, described respectively in the tables 7 and 8 as a sample data from the dataset :

- $c_{2,1}$: the bayesian probability of a vehicle is related to the clusters, as represented in the table 7
- $c_{2,2}$: the database sites are in the same clusters and the bayesian probability of a vehicle is related to different sites, as represented in the table 8.

Finally four tables are used in the database as specified in the tables 5, 6, 7 and 8 where all the dataset is constituted by duplicated vehicles names with its specification.

Table 5. Table of vehicles

Region name	Cluster Ref	Site Ref	Seller name	Vehicle name	Caracteristics	quantity
secteur 1	(0, 0)	(0,1)	seller 1	toyota avensis	vitesse automatique	40
secteur 1	(0,0)	(0,1)	seller 1	toyota avensis	vitesse automatique	40
secteur 1	(0,0)	(0,1)	seller 1	toyota avensis	vitesse automatique	40
secteur 1	(0,0)	(0,1)	seller 1	toyota avensis	vitesse automatique	40
secteur 1	(0,0)	(0,1)	seller 11	toyota avensis	vitesse automatique	40
–	–	–	–	–	–	–

Table 6. Table of regions

Region name	images
secteur 1	image blob
–	–

The following computer characteristics have been used for simulation of the proposed method :

- Processor : Intel® Core™ i7-4600M CPU @ 2.90GHz × 4

Table 7. the first memory table : $c_{2,1}$: the bayesian probability of a vehicle is related to the clusters

Cluster reference	Vehicle BP on the cluster	Vehicle
(0,0)	0.5	toyota avensis
(1,0)	0.5	toyota avensis
(0,1)	0.5	toyota avensis
(1,1)	0.5	toyota avensis
–	–	–

Table 8. the second memory table : $c_{2,2}$: the bayesian probability of a vehicle is related to the sites in the same cluster

Seller REF	Seller name	Vehicle	Vehicle BP on the seller
(0,1)	seller 1	toyota avensis	0.6
(0,1)	seller 1	toyota rav4	0.7
–	–	–	–

- Graphic card : Intel® HD Graphics 4600 (HSW GT2)
- Memory usable : 7.7 GB
- system : Ubuntu 20.04 LTS 64 bits
- Spyder as a python language editor, installed on localhost
- Mongodb as DBMS, installed on localhost
- spark as a data processing tool connected on mongodb to implement the map reduce pattern.

In Mongodb, there is also possible to use directly mapReduce through mongo shell, instead to consider integrated mapReduce in spark. That raises a question to compare comparisons between time complexity. Agregation in mongodb can also replace mapReduce. Agregation is defined by : But, here is an implementation of mapReduce, described in the section 3.3, in javascript respectively in figure x and y.

In Mongodb, the previous tables 5, 6, 7 and 8 correspond to the concept of collections. Each line in these tables is associated to a document in a collection as follows :

- the table 5 is associate to the collection in the table 9
- the table 6 becomes the collection represented in the table 10
- the table 7 is associated to the collection in the table 11
- the table 8 is represented by the collection in the table 12

The document 1, 2, 3, 4 in mongodb can be localized in different collection or in the same collection. The data has been repeated in the database to have enormous data in order to measure the impact of the bayesian probability : If the data is large, the role of an assistant that implements the bayesian approach will be more perceptible.

When a site is near, accordingly to a timetable planned in common with the other sites, synchronisation will be realize between its local database and the central database, to update data and the associated bayesian probability, to ensure data quality in the central database. Synchronisation is also executed through threads running in a background process, scheduled as a trigger in the database.

Table 9. A document in the collection

Document 1
{ regionname : "secteur1"
clusterref : "(0,0)"
siteref:"(0,1)"
sellername :"sellers1"
vehiclename : "toyota avensis"
caracteristics: "vitesse automatique"
quantity :40 }

Table 10. A document in the collection

Document 2
{ regionname : "secteur1"
images : image blob }

Table 11. A document in the collection

Document 3
{ clusterRef : "(0,0)"
vehicleBP : 0.5
vehiclename :"Toyota avensis" }

Table 12. A document in the collection

Document 4
{ sellerref : "(0,1)"
sellername : "seller 1"
vehiclename :"Toyota avensis"
vehicleBP : 0.5 }

The proposed solution takes the previous work using parallelism through the implementation of map-reduce framework by Abdoulaye sere and others. in [16] into account, but introduces bayesian probability to reduce the execution time. The composition of the bayesian approach and the parallel case appears naturaly better than the method that uses only the parallel case. Using only the parallel case considers that each cluster or each database site has the same probability : in this case, each cluster or each database site has the same priority in data research.

The proposed method are limitations, according to several contexts listed here:

- the time taken by the DBMS to udpate the table T_i in the background process does not influe really on the global time to find out data, requested by the users : in this way, the background process will be scheduled in a free period on the server, when the server is not in charge.

– scability is not considered on the DBMS, where several requests are launched on the server. Because in scability, others transactions in others thread can influe on the execution time of the current thread.

5 Conclusion and perspectives

Data research in a smart gis database has been analyzed with bayesian probability to predict how to realize data research. In this way, the nearest sites have been determined through mesh generation and the map-reduce algorithm to speed data processing, using bayesian approach to predict the data research. The role of Bayesian approach is to indicate the best probability for the clusters and the database sites, which will be concern firstly with data research, in order to avoid to lost time in founding data in research in the clusters or in the database sites.

It appears that a smart GIS database using bayesian approach is better than the method based on a virtual grid application and map reduce algorithm, presented by sere and others in [16] in term of efficiency : that means to consider finding an item in the GIS database in a short time.

Moreover, bayesian probability, substituted by others probabilities or to add new probabilities to the criteria defined in the list of constraints, might contribute to reduce the time taken in data research to obtain a smart database.

Instead of using the bayesian probability, others techniques in Artificial Intelligence, such as neural networks implemented in the components of the central DBMS might be studied to predict which clusters or sites is good for research : that needs to get more data for analysis.

In the future works, analysis of data research in a graph, precisely using a decision tree as a model, in the GIS database, implemented by the Nosql database based on a graph model for instance, will be studied to represent a road by a vertex and a data by a node.

References

1. Abdoulaye SERE, *Transformations analytiques appliquées aux images multi-échelles et bruitées*, thèse de doctorat unique en informatique, Université de Ouagadougou, 2013
2. Abdoulaye Sere and Dario Colazzo and Oumarou Sie, *A Hough Transform based on a Map-Reduce Algorithm*, International Journal of Engineering Research and Applications (IJERA), 2016
3. Jimmy Lin, *MapReduce Algorithm* Design, Tutorial, Rio de Janeiro, 2013
4. Jairam Chandar, *Join Algorithms using Map-Reduce*, Master of Science, Computer Science School of Informatics, 2010
5. Jesus Camacho-Rodriguez, Dario Colazzo, Ioana Manolescu, *PAXQuery : Efficient Parallel Processing of Complex XQuery*, IEEE, 2015
6. Zeba Khanam, Shafali Agarwa, *Map-Reduce implementations : survey and performance comparison*, International Journal of Computer Science and Information Technology (IJCSIT), volume 7 , issue 4, 2015.

7. J. Dean and S. Ghemawat, *Map-reduce: simplified data processing on large clusters*, Commun. ACM, volume 51, issue 1, page 107-113, 2008.
8. David COEURJOLLY, *algorithmique et géométrie discrète pour la caractérisation des courbes et des surfaces*, thèse de doctorat en informatique, université Lumière Lyon2, 2002.
9. Marie-Andree and Jacob-Da Col, *Applications quasi-affines et pavages du plan discret*, Theoretical Computer Science vol 259 page 245-269, 2001
10. Zhi-Qiang Feng, Zhengqun Guan, Zhuowei Chen. *FER/Mesh : un logiciel de génération automatique de maillages*, 9e Colloque national en calcul des structures, CSMA, May 2009, Giens, France. hal- 01413781
11. Vincent François, *Méthodes de maillage et de remaillage automatiques appliquées à la modification de modèle dans le contexte de l'ingenierie simultanée*, thèse de doctorat, Université Henri Poincaré - Nancy 1, 1998
12. Abdoulaye Sere, José Arthur Ouedraogo, Boureima Zerbo, Oumarou Sie, *Post Classification in the Social Networks using the Map-reduce Algorithm*, (IJACSA) International Journal of Advanced Computer Science and Applications,Vol. 11, No. 12, 2020
13. Mateus Coelho ; Dylan Sugimoto ; Gabriel Melo ; Vitor Curtis and Juliana Bezerra, *A MapReduce based Approach for Circle Detection*, In Proceedings of the 14th International Conference on Software Technologies - Volume 1: ICSOFT, 454-459, Prague, Czech Republic, 2019.
14. Irina Astrova, Arne Koschel, Nils Wellermann, Philip Klostermeyer, *Performance Benchmarking of NewSQL Databases with Yahoo Cloud Serving Benchmark*, Springer Nature Switzerland AG 2021, K. Arai et al. (Eds.): FTC 2020, AISC 1289, pp. 271–281, 2021.
15. Moubaric Kabore, *Application du framework map-reduce à des groupes de données massives*, mémoire de master en Informatique, Université Joseph KI-ZERBO, 2017.
16. Abdoulaye Sere, Jean Serge Dimitri Ouattara, Didier Bassole, Jose Arthur Ouedraogo and Moubaric Kabore, *A Framework for Data Research in GIS Database using Meshing Techniques and the Map-Reduce Algorithm*, International Journal of Advanced Computer Science and Applications(IJACSA), 12(3), 2021. http://dx.doi.org/10.14569/IJACSA.2021.0120374
17. Bassole D., Ouedraogo F.T., Sie O. (2018) *G-Cloud: Opportunities and Security Challenges for Burkina Faso*, In: Bissyande T., Sie O. (eds) e-Infrastructure and e-Services for Developing Countries. AFRICOMM 2016. Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, vol 208. Springer, Cham.
18. Yue Dong, A. Goh, *An intelligent database for engineering applications*, Artificial Intelligence in Engineering, Volume 12, Issues 1–2, 1998, Pages 1-14, ISSN 0954-1810, https://doi.org/10.1016/S0954-1810(96)00033-7.
19. X. Zhou, C. Chai, G. Li and J. SUN, *Database Meets Artificial Intelligence: A Survey*, in IEEE Transactions on Knowledge and Data Engineering, doi: 10.1109/TKDE.2020.2994641.
20. Huang, Junpeng, Sixiang Ling, Xiyong Wu, and Rui Deng. 2022. *GIS-Based Comparative Study of the Bayesian Network, Decision Table, Radial Basis Function Network and Stochastic Gradient Descent for the Spatial Prediction of Landslide Susceptibility*, Land 11, no. 3: 436. https://doi.org/10.3390/land11030436
21. C. Ordóñez Galán, J.M. Matías, T. Rivas, F.G. Bastante, *Reforestation planning using Bayesian networks*, Environmental Modelling and Software, Volume 24, Issue 11, 2009, Pages 1285-1292, ISSN 1364-8152, https://doi.org/10.1016/j.envsoft.2009.05.009.

22. A. Stassopoulou, M. Petrou, J. Kittler (1998), *Application of a Bayesian network in a GIS based decision making system*, International Journal of Geographical In formation Science, 12:1, 23-46, DOI: 10.1080/136588198241996.
23. Ames, D., Anselmo, A. (2008). *Bayesian Network Integration with GIS*, in Shekhar, S., Xiong, H. (eds) Encyclopedia of GIS. Springer, Boston, MA. https://doi.org/10.1007/978-0-387-35973-1-95.
24. Antoine Vacavant, *Géométrie discrète sur grilles irrégulières isothétiques*, thèse de doctorat en informatique, Université Lumière Lyon 2, 2007,